Image Stitching Report

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Description

This assignment involved creating a panorama by stitching two images together using homography transformations. The process required detecting key points with SIFT, matching them, and getting the homography matrix with RANSAC to align the images. The final step was mapping pixels from one image to the other using the inverse homography.

Image Stitching Process

The process began by loading the images and detecting key points with SIFT. After matching the points, the homography matrix was calculated using RANSAC for outliers. The second image was then aligned with the first using the inverse of the homography matrix, and the pixels were copied onto the panorama.

Observations

The stitching algorithm performed well for images with distinct features like cityscapes and landscapes, where key points were easy to find. For images with low contrast or repeating patterns, the algorithm had trouble finding matches which led to misalignment in the panorama. Images with changing scales or angles caused issues, especially when the overlap between the images was small, making it difficult to align them.

Difficulties

The main difficulty was making sure the images aligned correctly when there weren't enough good key points to match, like in low contrast images. Calculating the homography matrix and copying pixels correctly using the inverse transformation was tricky when dealing with areas outside the overlap. I also had to adjust the boundaries of the panorama to avoid misplacement of the images.

Images

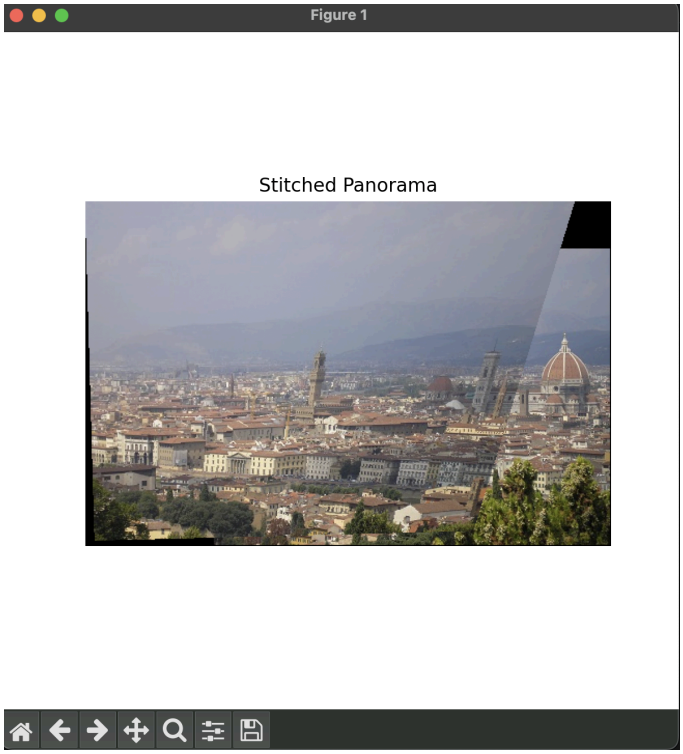


Fig. 1

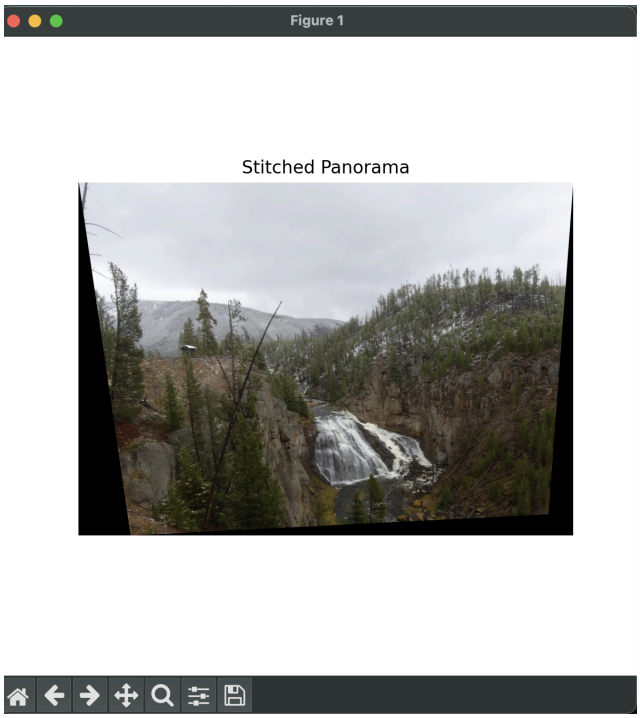


Fig.2

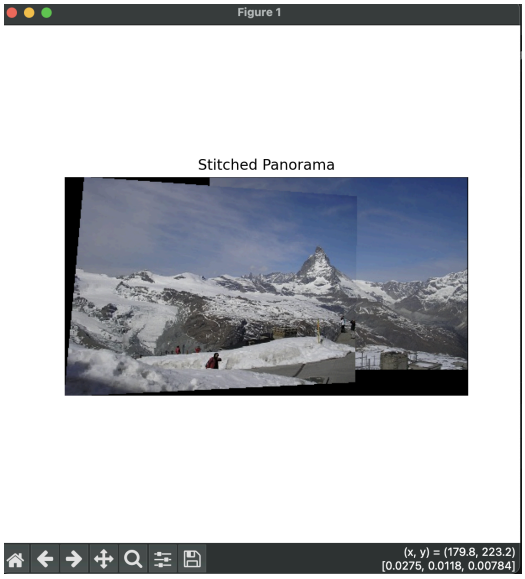

Figure 1
Stitched Panorama
(x, y) = (179.8, 223.2)
[0.0275, 0.0118, 0.00784]

Fig.3

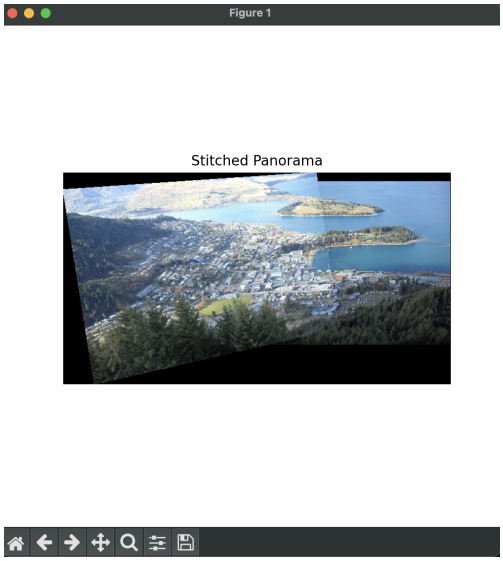

Figure 1
Stitched Panorama

Fig.4

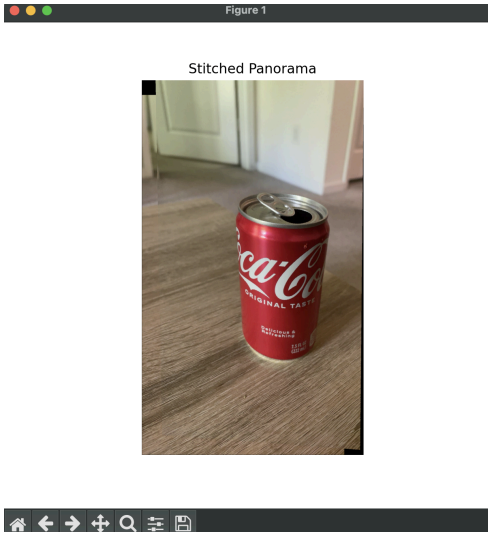


Fig.5